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PRST endsem solution

Question 1 of 5

a)

$n = 10$

$\bar{x} = 60.8$ $\quad var = 42.56 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

$= \frac{1}{n}\sum x_i^2 - (\bar{x})^2$

58 replaced by 50

67 " by ~~67~~ 65

new total $= 60.8 \times 10 - 8 - 2$

$= 608 - 10$

$= 598$

(3)

new mean $\underline{\bar{x}_{new}} = \frac{598}{10} = \boxed{59.8}$

old ~~new~~ sum of square $\sum_{i=1}^{10} x_i^2 = 10\left[42.56 + (60.8)^2\right]$

$= 37392$

$\Rightarrow \sum_{i=1}^{8} x_i^2 + 67^2 + 58^2 = 37392$

Then $\sum_{i=1}^{8} x_i^2 + 65^2 + 50^2 = 37392 - 67^2 - 58^2 + 65^2 + 50^2$

$= \cancel{45242}\ 36264$

new sum of square $= \cancel{45242}\ 36264$

$\underline{\text{new var}} = \frac{\cancel{45242}\ 36264}{10} - (\bar{x}_{new})^2$

(5)

$= \boxed{50.36}$

b) $\frac{1}{n}\Sigma(x_i - 1) = 2.6 = m_1'$

$\frac{1}{n}\Sigma(x_i - 1)^2 = 10.2 = m_2'$

$\frac{1}{n}\Sigma(x_i - 1)^3 = 43.4 = m_3'$

$\frac{1}{n}\Sigma(x_i - 1)^4 = 192.6 = m_4'$

then $\bar{x} = \frac{1}{n}\Sigma x_i = 2.6 + 1 = 3.6$

Using relatn between central and non central moments

where $d = \bar{x} - A$
$= \bar{x} - 1$
$= 3.6 - 1$
$= 2.6$

$m_1 = m_1' - d$
$= 2.6 - (3.6 - 1)$
$= 0$

$m_2 = m_2' - 2m_1'd + d^2$
$= 10.2 - 2(2.6)\times(2.6) + (2.6)^2$
$= 3.44$ (1)

$m_3 = m_3' - 3m_2'd + 3m_1'd^2 - d^3$
$= 43.4 - 3\times(10.2)(2.6) + 3.(2.6)(2.6)^2 - (2.6)^3$
$= -1.008$ (1)

~~$m_4 = m_4'$~~

$m_4 = m_4' - 4m_3'd + 6m_2'd^2 - 4m_1'd^3 + d^4$
$= 192.6 - 4(43.4)(2.6) + 6(10.2)(2.6)^2 - 4\times(2.6)(2.6)^3 + (2.6)^4$
$= 17.8592$ (1)

$\beta_1 = \dfrac{m_3^2}{m_2^3} = 0.0249$ (~~3~~) (~~1~~)

$\gamma_1 = \dfrac{m_3}{m_2^{3/2}} = -0.1579$

$\beta_2 = \dfrac{m_4}{m_2^2} = 1.509$ (~~2~~) (1)

$\gamma_2 = \beta_2 - 3 = -1.490$

as $\gamma_1 < 0$ the dist is -vely skew — (1)

$\gamma_2 < 0$ the dist is platykurtic (1)

$P(Y=3) = 0.5$ (1)

c) $X = 2, 4$

$P(X=2 \mid Y=3) = \dfrac{P(X=2, Y=3)}{P(Y=3)} = \dfrac{0.20}{0.20+0.30} = \dfrac{0.2}{0.5} = \dfrac{2}{5}$ (1)

$P(X=4 \mid Y=3) = \dfrac{P(X=4, Y=3)}{P(Y=3)}$ (1)

$= \dfrac{0.3}{0.5} = \dfrac{3}{5}$

$E(X \mid Y=3) = 2 \times \dfrac{2}{5} + 4 \times \dfrac{3}{5}$ (2)

$= \dfrac{4+12}{5} = \dfrac{16}{5}$